Adriano Guastella

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Professional Experience

Machine Learning Engineer, Department of Computer Science and Engineering (DISI), University of Bologna, Italy — June 2024 – Jan 2026

- Engineered production-ready federated learning systems for resource-constrained edge devices, focusing on model sparsity and distributed optimization
- Researched federated unlearning techniques to enable selective data removal from trained models while preserving privacy and model performance

Visiting Researcher, CaMLSys Lab, University of Cambridge, England — Oct 2023 – Feb 2024

- Built prototypes for federated learning systems using PyTorch and the Flower framework, integrating Powerpropagation into a sparse-training pipeline
- Developed a comprehensive experimental framework to benchmark sparse training methods, enabling systematic performance evaluation across diverse datasets

Software Engineer Intern, IIT-CNR of Pisa, Italy — Apr 2021 – June 2021

- Architected scalable Twitter data collection system enabling historical tweet retrieval through custom REST API
- Built end-to-end data processing pipelines on Apache NiFi for production social listening platform, integrated as module in a broader web data collector

Technical Skills

ML & Tooling: PyTorch, Flower, Hugging Face, Weights & Biases, NumPy, Pandas, scikit-learn
Languages: Python, C/C++, Go, JavaScript, SQL
Systems & DevOps: Linux, Docker, Git, Slurm, Apache NiFi, Elasticsearch, MySQL
Domain: Federated Learning, Privacy-Preserving ML, ETL Pipelines

Education

University of Bologna, M.S. in Computer Engineering, 110/110 *cum laude* — 2021 – 2024

University of Pisa, Bachelor's degree in Computer Engineering, 91/110 — 2015 – 2021

Publications & Conferences

Sparsity in Federated Learning: A Survey — August 2026

A. Mora, **A. Guastella**, L. Sani, P. Bellavista, N. D. Lane
Accepted at International Joint Conference on Artificial Intelligence (IJCAI-ECAI) 2026, Survey Track, Bremen, Germany

SparsyFed: Sparse Adaptive Federated Training — April 2025

A. Guastella*, L. Sani*, A. Iacob, A. Mora, P. Bellavista, N. D. Lane (*equal contribution) arXiv:2504.05153
Accepted at International Conference on Learning Representations (ICLR) 2025, Singapore
Trains neural networks at 95% sparsity with limited accuracy drop, achieving 19× communication reduction vs. dense baselines for cross-device FL deployments.

Languages

Italian: Native **English:** C1 level (IELTS Academic 7.0) **Spanish:** B1 level